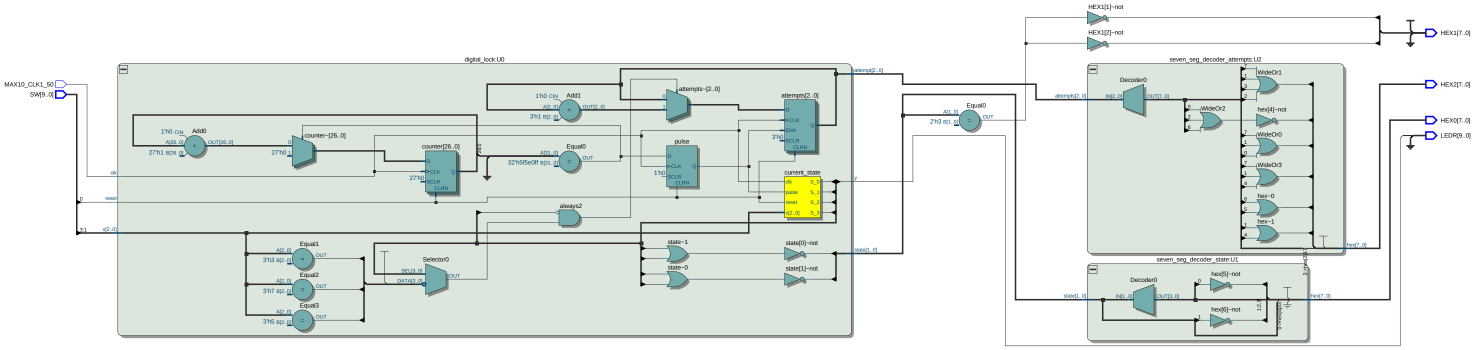
MAX10_CLK1_50
SW[9..0]
clk
0
reset
3:1
x[2..0]
digital_lock:U0
1'h0 CIN
Add0
A[26..0]
OUT[26..0]
27'h1 B[26..0]
counter~[26..0]
0
27'h0 1
counter[26..0]
D
CLK
Q
27'h0
SCLR
CLRN
26:0
1'h0 CIN
Add1
A[2..0]
OUT[2..0]
3'h1 B[2..0]
attempts~[2..0]
0
1
attempts[2..0]
D
CLK
ENA
Q
3'h0
SCLR
CLRN
Equal0
A[31..0]
OUT
32'h5f5e0ff B[31..0]
pulse
D
CLK
Q
1'h0
SCLR
CLRN
current_state
clk
pulse
reset
x[2..0]
S_0
S_1
S_2
S_3
y
always2
Equal1
A[2..0]
OUT
3'h3 B[2..0]
Equal2
A[2..0]
OUT
3'h7 B[2..0]
Equal3
A[2..0]
OUT
3'h5 B[2..0]
Selector0
SEL[3..0]
DATA[3..0]
OUT
state~1
state~0
state[0]~not
state[1]~not
attempt[2..0]
state[1..0]
Equal0
A[1..0]
OUT
2'h3 B[1..0]
HEX1[1]~not
HEX1[2]~not
HEX1[7..0]
seven_seg_decoder_attempts:U2
attempts[2..0]
Decoder0
IN[2..0]
OUT[7..0]
WideOr1
7
1
3
2
WideOr2
0
2
6
hex[4]~not
WideOr0
7
1
0
WideOr3
7
1
4
hex~0
6
5
hex~1
1
4
hex[7..0]
2->hex[2]1:7
HEX2[7..0]
seven_seg_decoder_state:U1
state[1..0]
Decoder0
IN[1..0]
OUT[3..0]
hex[5]~not
0
hex[6]~not
1
1:2,1
0->hex[4]3:3
hex[7..0]
HEX0[7..0]
LEDR[9..0]